



AyuAstro

NOTHING TO HIDE

Complete Technical Documentation

Algorithm Design, Architecture, Implementation Details & System
Specification

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Stack: Next.js 16 · TypeScript · Swiss Ephemeris · Prisma · Tailwind CSS

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1 Product Vision & Philosophy

AyuAstro is an AI-powered Vedic astrology platform that provides brutally honest, psychologically grounded emotional intelligence reports. Founded by Ayush UPADHYAY, the platform's tagline — *"Nothing to Hide"* — reflects its core philosophy: no sugarcoating, no Barnum statements, no superstition.

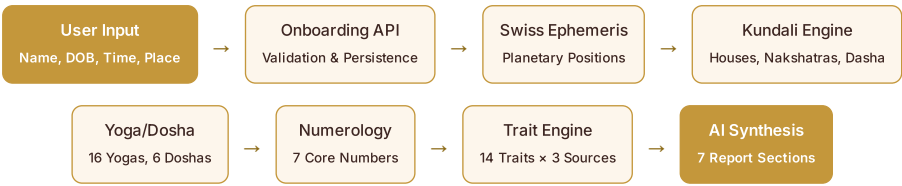
Core Principles

- Deterministic Calculations:** Same birth details → identical results, always. No randomness, no AI hallucination in calculations.
- Swiss Ephemeris Accuracy:** Professional-grade arc-minute level planetary positions using the same library trusted by professional astrologers worldwide.
- Psychological Grounding:** Every interpretation is anchored in planetary dignity, house placement, and classical Vedic rules — never vague fortune-telling.
- Multi-Source Synthesis:** Astrology (40%) + Numerology (20%) + Behavioral Science (40%) = holistic emotional architecture mapping.
- AI as Synthesizer Only:** AI never defines truth — it only synthesizes and articulates what the deterministic algorithms have computed.

Target Users

Spiritually curious individuals (20-40 years old, primarily Indian market) seeking self-understanding through the lens of Vedic wisdom, combined with modern psychological insight.

2 Architecture Overview



Layer Architecture

Layer	Technology	Purpose
Frontend	Next.js 16 + React 19 + Tailwind CSS 4	SPA with 8+ views, shadcn/ui components, Framer Motion
API Layer	Next.js App Router + Zod	RESTful endpoints with strict validation
Astrology Engine	TypeScript + Swiss Ephemeris (sweph)	Deterministic planetary calculations
Numerology Engine	TypeScript (Pythagorean system)	7 numerological calculations
Scoring Engine	TypeScript (rule-based)	14 traits scored 0-100 from 3 sources
AI Layer	z-ai-web-dev-sdk (LLM)	Report synthesis, chat, horoscope
Database	Prisma ORM + SQLite	Persistent storage for user data

3 Technology Stack

Component	Library / Tool	Version
Framework	Next.js (App Router)	16.x
Language	TypeScript	5.x
Styling	Tailwind CSS + shadcn/ui	4.x / New York
Animations	Framer Motion	11.x
State	Zustand (persisted)	5.x
Ephemeris	sweeph (Swiss Ephemeris N-API)	2.10.3
Database	Prisma + SQLite	6.x
Validation	Zod	3.x
AI SDK	z-ai-web-dev-sdk	Latest
Icons	Lucide React	Latest

4 Data Flow Pipeline

The complete processing pipeline transforms raw birth data into a comprehensive emotional intelligence report through 6 sequential stages:

Stage	Input	Output	Time
1. User Creation	Name, DOB, Time, Place, Gender, Relationship	userId + Profile in DB	~50ms
2. Astrology Calculation	DOB, Time, Lat/Lng, Timezone	Sun/Moon/Asc signs, 9 planet positions, 12 houses, Nakshatra, Dasha, 16 Yogas, 6 Doshas	~200ms
3. Numerology Calculation	Full Name + DOB	7 core numbers with descriptions	~10ms
4. Trait Scoring	Astrology + Numerology + Questionnaire	14 traits scored 0-100 with audit trail	~50ms
5. AI Report Generation	All computed data + templates	7 report sections (3 free + 4 premium)	~12s
6. Persistence	All results	Complete data saved to DB + Zustand store	~50ms

Total Pipeline Time: ~13 seconds (dominated by AI generation). All deterministic calculations (Stages 1-4) complete in under 500ms.

5 Swiss Ephemeris Integration

The **primary calculation path** uses the Swiss Ephemeris (`sweph`) — the same professional-grade astronomical library used by astrologers and researchers worldwide. It provides **arc-minute level accuracy** (vs ~1-3° error in simplified methods).

Module Architecture

```
// swiss-ephemeris.ts - Wrapper Module

// Lazy initialization with graceful fallback
let swephModule: typeof import('sweph') | null = null;

async function initSweph(): Promise<{ ready: boolean; error?: string }> {
  try {
    swephModule = require('sweph');
    // Test with utc_to_jd to verify module works
    const testResult = swephModule.utc_to_jd(2000, 1, 1, 12, 0, 0,
      swephModule.constants.SE_GREG_CAL);
    return { ready: true, version: '2.10.3' };
  } catch (err) {
    swephModule = null;
    return { ready: false, error: err.message };
  }
}
```

Planet ID Mapping

Planet	Swiss Ephemeris Constant	Notes
Sun ☉	SE_SUN	Central luminary
Moon ☾	SE_MOON	Lunar position
Mars ♂	SE_MARS	
Mercury ☿	SE_MERCURY	
Jupiter ♃	SE_JUPITER	
Venus ♀	SE_VENUS	
Saturn ♄	SE_SATURN	
Rahu ♃	SE_MEAN_NODE	Vedic standard: mean node
Ketu ♄	SE_MEAN_NODE + 180°	Derived from Rahu

Calculation Flags

```
const flags = constants.SEFLG_MOSEPH | constants.SEFLG_SPEED;
// SEFLG_MOSEPH → Use built-in Moshier ephemeris (no external files needed)
// SEFLG_SPEED → Calculate longitudinal speed (negative = retrograde)
```

House System

```
// Whole Sign house system ('W') - Vedic standard
const result = swephModule.houses_ex2(jd_ut, flags, latitude, longitude, 'W');
```

Lahiri Ayanamsa

```
// Set Lahiri sidereal mode before getting ayanamsa
swephModule.set_sid_mode(constants.SE_SIDM_LAHIRI, 0, 0);
const ayanamsa = swephModule.get_ayanamsa_ut(jd_ut, 0);
```

Important: The sweeph module uses native N-API bindings and can ONLY be used in server-side code (API routes, server components). It cannot be imported in client-side code.

6 Meeus Fallback Engine

When Swiss Ephemeris is unavailable (e.g., N-API module fails to load in certain runtime environments), the system automatically falls back to simplified **Meeus orbital mechanics** with $\sim 1\text{--}3^\circ$ accuracy. This ensures the application never crashes due to ephemeris issues.

Planetary Longitude Formulas (Simplified)

Each planet's tropical longitude is calculated using the Meeus method:

```
// Sun Longitude (Meeus)
Ls = 280.46646 + 36000.76983 × T + 0.0003032 × T2 // Mean longitude
M = 357.52911 + 35999.05029 × T - 0.0001537 × T2 // Mean anomaly
C = (1.914602 - 0.004817×T) × sin(M) // Equation of center
  + 0.019993 × sin(2M) + 0.000289 × sin(3M)
SunLongitude = Ls + C

// Moon Longitude (Meeus)
L = 218.3165 + 481267.8813 × T // Mean longitude
M = 134.9634 + 477198.8676 × T // Mean anomaly
Ms = 357.5291 + 35999.0503 × T // Sun's mean anomaly
D = 297.8502 + 445267.1115 × T // Mean elongation
F = 93.2720 + 483202.0175 × T // Argument of latitude
MoonLongitude = L + 6.289×sin(M) + 1.274×sin(2D-M) + ...
```

Retrograde Detection (Meeus)

Without Swiss Ephemeris speed data, retrograde is approximated from elongation angles:

```
function isPlanetRetrograde(planet, tropicalLon, sunTropicalLon) {
  const elongation = normalize(tropicalLon - sunTropicalLon);
  switch (planet) {
    case 'Mercury': return elongation < 22 || elongation > 338;
    case 'Venus':   return elongation < 10 || elongation > 350;
    case 'Mars':    return elongation > 130 && elongation < 230;
    case 'Jupiter': return elongation > 145 && elongation < 215;
    case 'Saturn':  return elongation > 150 && elongation < 210;
  }
}
```

7 Julian Day Calculation

The Julian Day Number (JDN) is the foundational time coordinate for all astronomical calculations. Two variants are used:

- `jd_et` — Ephemeris Time (Terrestrial Time): Used for planet position calculations
- `jd_ut` — Universal Time: Used for house cusps, ascendant, and ayanamsa

Meeus Algorithm

```
function dateToJulianDayMeeus(date: Date): number {
  let y = year, m = month;
  if (m ≤ 2) { y -= 1; m += 12; }
  const A = Math.floor(y / 100);
  const B = 2 - A + Math.floor(A / 4);
  return Math.floor(365.25 × (y + 4716))
    + Math.floor(30.6001 × (m + 1))
    + day + hour/24 + B - 1524.5;
}

// Julian centuries from J2000.0
function julianCenturies(jd: number): number {
  return (jd - 2451545.0) / 36525.0;
}
```

8 Lahiri Ayanamsa

The **ayanamsa** is the angular difference between the tropical (sayana) and sidereal (nirayana) zodiacs. AyuAstro uses the **Lahiri (Chitrapaksha)** ayanamsa — the official standard for Vedic astrology in India.

Swiss Ephemeris Method (Primary)

```
set_sid_mode(SE_SIDM_LAHIRI, 0, 0);
const ayanamsa = get_ayanamsa_ut(jd_ut, 0);
```

Meeus Fallback Formula

```
function calculateLahiriAyanamsaMeeus(date: Date): number {
  const T = julianCenturies(jd);
  const omega = 259.18 - 1934.14 × T;
  return 23.85394444 + 0.01396222 × (T × 100)
    + 0.00156 × sin(omega × π/180);
}
```

Current Value: As of 2025, Lahiri ayanamsa ≈ 24°12' (24.2°). This means a tropical Aries at 25° becomes a sidereal Aries at ~0.8° — barely in Aries. All AyuAstro calculations use sidereal positions.

9 Planetary Position Calculation

For each of the 9 Vedic planets, the following data is computed:

PlanetPosition Interface

```
interface PlanetPosition {
  planet: Planet;           // 'Sun' | 'Moon' | 'Mars' | 'Mercury' | 'Jupiter' | 'Venus' | 'Saturn' | 'Rahu' | 'Ketu'
  longitude: number;        // 0-360 tropical longitude
  siderealLongitude: number; // 0-360 sidereal (Lahiri) longitude
  sign: ZodiacSign;         // Which zodiac sign the planet occupies
  signIndex: number;        // 0-11 (0=Aries, 11=Pisces)
  degreeInSign: number;     // 0-30 degrees within the sign
  house: number;            // 1-12 house number (whole sign system)
  nakshatra: string;        // Nakshatra name
  nakshatraIndex: number;   // 0-26
  nakshatraPada: number;    // 1-4
  isRetrograde: boolean;    // True if planet is moving backward
  isCombust: boolean;       // True if too close to Sun
  eclipticLatitude?: number; // From Swiss Ephemeris
  speed?: number;           // Degrees/day (negative = retrograde)
  distanceAU?: number;      // Distance from Earth in AU
}
```

Conversion Process

```
// Step 1: Get tropical longitude from Swiss Ephemeris
const tropicalLon = swephCalcPlanet(jd_et, planet).longitude;

// Step 2: Convert to sidereal by subtracting ayanamsa
const siderealLon = normalize(tropicalLon - ayanamsa);

// Step 3: Derive sign from sidereal longitude
const signIndex = Math.floor(siderealLon / 30);
const sign = ZODIAC_SIGNS[signIndex]; // e.g., 'Aries', 'Taurus', ...
const degreeInSign = siderealLon % 30;

// Step 4: Derive nakshatra from sidereal longitude
const nakshatraIndex = Math.floor(siderealLon / 13.333); // 360/27
const nakshatraPada = Math.floor((siderealLon % 13.333) / 3.333) + 1;

// Step 5: Retrograde from actual speed (Swiss Ephemeris)
const isRetrograde = (planet === 'Rahu' || planet === 'Ketu')
  ? true // Nodes always retrograde in Vedic astrology
  : lonSpeed < 0;

// Step 6: Combustion check
const isCombust = angularDistance(planetLon, sunLon) < COMBUSTION_DEGREES[planet];
```


10 Ascendant (Lagna) Calculation

The Ascendant is the zodiac sign rising on the eastern horizon at the moment of birth. It forms the first house of the kundali and is critical for all house-based interpretations.

Swiss Ephemeris Method

```
const houseData = swephCalcHouses(jd_ut, latitude, longitude);
const ascTropical = houseData.ascendant;
const ascSidereal = normalize(ascTropical - ayanamsa);
const ascSignIndex = Math.floor(ascSidereal / 30);
```

Meeus Fallback Method

```
// Local Sidereal Time → Ramc → Ascendant
const gmst = 280.46061837 + 360.98564736629 * (jd - 2451545.0) + 0.000387933 * T2;
const lst = normalize(gmst + longitude);
const ramc = lst * π/180;
const obliquity = (23.4393 - 0.0130 * T) * π/180;
const ascendant = atan2(cos(ramc),
  -(sin(ramc) * cos(obliquity) + tan(latitude) * sin(obliquity)));
```

11 Whole Sign House System

AyuAstro uses the **Whole Sign house system** — the oldest and most traditional Vedic system. In this system, each zodiac sign equals one house, starting from the ascendant sign as House 1.

Algorithm

```
function calculateHouses(ascendant, planetaryPositions) {
  const houses = [];
  for (let houseNum = 1; houseNum ≤ 12; houseNum++) {
    const signIndex = (ascendant.signIndex + houseNum - 1) % 12;
    const sign = ZODIAC_SIGNS[signIndex];

    // Find planets in this house (same sign = same house in whole sign)
    const planetsInHouse = Object.entries(planetaryPositions)
      .filter(([_, pos]) => pos.signIndex === signIndex)
      .map(([name]) => name);

    houses.push({ houseNumber: houseNum, sign, signIndex, planets: planetsInHouse });
  }
  return houses;
}
```

12 House Significations

House	Sanskrit	Name	Significations	Karaka
1	Lagna / Tanu	Self	Body, Personality, Health	Sun
2	Dhana	Wealth	Money, Family, Speech	Jupiter
3	Sahaja	Siblings	Courage, Communication	Mars
4	Sukha	Home	Mother, Comforts, Education	Moon
5	Putra	Children	Intelligence, Creativity, Romance	Jupiter
6	Ripu	Enemies	Disease, Debt, Service	Saturn
7	Kalatra	Partnership	Marriage, Business Partners	Venus
8	Ayur	Transformation	Longevity, Inheritance, Occult	Saturn
9	Bhagya	Fortune	Luck, Religion, Father, Dharma	Jupiter
10	Karma	Career	Status, Authority, Public Image	Sun
11	Labha	Gains	Income, Friends, Social Circle	Jupiter
12	Vyaya	Loss	Expenses, Liberation, Foreign	Saturn

12 Nakshatra Calculation

The 27 Nakshatras (lunar mansions) divide the zodiac into 27 equal segments of 13°20' each. Each nakshatra has 4 padas (quarters) of 3°20'.

```
// Constants
DEGREES_PER_NAKSHATRA = 360 / 27 = 13.333... // 13°20'
DEGREES_PER_PADA = DEGREES_PER_NAKSHATRA / 4 = 3.333... // 3°20'

// Calculate from sidereal longitude
const nakshatraIndex = Math.floor(siderealLongitude / 13.333);
const pada = Math.floor((siderealLongitude % 13.333) / 3.333) + 1;
```

Each nakshatra is ruled by one of the 9 planets, which determines the starting point of the Vimshottari Dasha system.

13 Vimshottari Dasha System

The Vimshottari Dasha is the most widely used planetary period system in Vedic astrology, based on the Moon's nakshatra at birth. The total cycle spans **120 years**.

Dasha Sequence & Durations

#	Planet	Years	Proportion
1	Ketu ☾	7	5.83%
2	Venus ♀	20	16.67%
3	Sun ☉	6	5.00%
4	Moon ☾	10	8.33%
5	Mars ♂	7	5.83%
6	Rahu ☊	18	15.00%
7	Jupiter ♃	16	13.33%
8	Saturn ♄	19	15.83%
9	Mercury ☿	17	14.17%
Total		120	100%

Algorithm

```
// Step 1: Find Moon's nakshatra lord = starting dasha planet
const startLord = NAKSHATRAS[moonNakshatraIndex].ruler;

// Step 2: Calculate fraction of nakshatra already traversed
const fractionElapsed = (moonDegreeInNakshatra) / 13.333;
const fractionRemaining = 1 - fractionElapsed;

// Step 3: First mahadasha has reduced duration
const firstDuration = DASHA_YEARS[startLord] * fractionRemaining;

// Step 4: Subsequent dashas follow standard sequence
for (let i = 1; i < 9; i++) {
  const planet = DASHA_SEQUENCE[(startIndex + i) % 9];
  const duration = DASHA_YEARS[planet];
  // Each mahadasha has 9 antardashas
}

// Step 5: Antardasha within each Mahadasha
antardashaYears = (mahadashaYears * antardashaPlanetYears) / 120;
```

14 Planet-Sign Relationships

Per Brihat Parashara Hora Shastra (BPHS), a planet's relationship with a sign is determined by its relationship with the **lord of that sign**:

Algorithm

```
function getPlanetSignRelationship(planet, sign) {
  if (getSignLord(sign) === planet) return 'Own Sign';
  const signLord = getSignLord(sign);
  const relationship = getPermanentRelationship(planet, signLord);
  switch (relationship) {
    case 'Friend':      return 'Friendly';
    case 'Enemy':       return 'Enemy';
    case 'Great Friend': return 'Friendly';
    case 'Great Enemy':  return 'Enemy';
    default:            return 'Neutral';
  }
}
```

Permanent Friendship Table

Planet	Friends	Enemies	Neutral
Sun ☉	Moon, Mars, Jupiter	Venus, Saturn	Mercury
Moon ☾	Sun, Mercury	Rahu, Ketu	Mars, Jupiter, Venus, Saturn
Mars ♂	Sun, Moon, Jupiter	Mercury	Venus, Saturn
Mercury ☿	Sun, Venus	Moon	Mars, Jupiter, Saturn
Jupiter ♃	Sun, Moon, Mars	Mercury, Venus	Saturn
Venus ♀	Mercury, Saturn	Sun, Moon	Mars, Jupiter
Saturn ♄	Mercury, Venus	Sun, Moon, Mars	Jupiter

Key Rule: Mars is *Neutral* to Mercury (NOT enemy), per BPHS. This was a common source of errors in earlier implementations where Mars was incorrectly listed as Mercury's enemy.

15 Exaltation & Debilitation

Each planet has a sign of **exaltation** (maximum strength) and a sign of **debilitation** (minimum strength), which is always the opposite sign (7th from exaltation).

Planet	Exaltation Sign	Exaltation Degree	Debilitation Sign
Sun ☉	Aries ♈	10°	Libra ♎
Moon ☾	Taurus ♉	3°	Scorpio ♏
Mars ♂	Capricorn ♐	28°	Cancer ♋
Mercury ☿	Virgo ♍	15°	Pisces ♛
Jupiter ♃	Cancer ♋	5°	Capricorn ♐
Venus ♀	Pisces ♛	27°	Virgo ♍
Saturn ♄	Libra ♎	20°	Aries ♈
Rahu ☊	Gemini ♊	15°	Sagittarius ♐
Ketu ☋	Sagittarius ♐	15°	Gemini ♊

17 Retrograde Detection

Swiss Ephemeris Method (Primary)

The actual longitudinal speed is extracted from the ephemeris.
Negative speed = retrograde.

```
const retrograde = (planet === 'Rahu' || planet === 'Ketu')  
  ? true // Nodes always retrograde in Vedic astrology  
  : lonSpeed < 0;
```

16 Meeus Approximation (Fallback)

Without speed data, retrograde is estimated from elongation angles:

Planet	Retrograde Zone (elongation from Sun)
Mercury	< 22° or > 338°
Venus	< 10° or > 350°
Mars	130° — 230°
Jupiter	145° — 215°
Saturn	150° — 210°

18 16 Yoga Detection Algorithms

Yogas are special planetary combinations indicating specific life patterns. AyuAstro detects 16 yogas:

#	Yoga	Detection Logic	Strength
1	Raj Yoga	Kendra lord in Trikona or vice versa, or conjunction of Kendra + Trikona lords	Strong if direct placement
2	Gaj Kesari	Jupiter in Kendra (1,4,7,10) from Moon	Strong if Jupiter in own/exalted sign
3	Neech Bhang Raj	Debilitated planet's dispositor in Kendra from Lagna/Moon	Strong if dispositor also strong
4	Chandra Mangal	Moon and Mars in conjunction or mutual aspect	Strong if conjunction
5	Budh Aditya	Sun + Mercury conjunction (not if Mercury combust)	Strong if Mercury not debilitated
6	Hansa	Jupiter in own/exalted/Moolatrikona sign in Kendra	Strong if exalted
7	Malavya	Venus in own/exalted/Moolatrikona sign in Kendra	Strong if exalted
8	Shasha	Saturn in own/exalted/Moolatrikona sign in Kendra	Strong if exalted
9	Ruchaka	Mars in own/exalted/Moolatrikona sign in Kendra	Strong if exalted
10	Bhadra	Mercury in own/exalted/Moolatrikona sign in Kendra	Strong if exalted
11	Amala	Venus AND Jupiter both in Kendra from Moon	Strong if both in own sign
12	Veshi	Planets (not Moon) in 2nd from Sun	3+ planets = Strong
13	Voshi	Planets (not Moon) in 12th from Sun	3+ planets = Strong
14	Ubhayachari	Planets in BOTH 2nd AND 12th from Sun	4+ planets = Strong
15	Dhana	2nd + 11th lords in conjunction or mutual aspect	Strong if direct conjunction
16	Vipreet Raj	Dushtana lords (6,8,12) in other Dushtana houses	Harsha + Sarala +
